



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Computer Science and Engineering (Data Science)

Lab Manual

Subject: Foundations of Data Analysis Laboratory (DJ19DSL303)

Semester: III

Experiment 6

(Data Visualisation)

Aim: Create an interactive data story.

Theory: In Tableau, a story is a sequence of visualizations that work together to convey information. You can create stories to tell a data narrative, provide context, demonstrate how decisions relate to outcomes, or to simply make a compelling case.

A story is a sheet, so the methods you use to create, name, and manage worksheets and dashboards also apply to stories (for more details, see Workbooks and Sheets). At the same time, a story is also a collection of sheets, arranged in a sequence. Each individual sheet in a story is called a story point.

“Statistics are rarely meaningful in and of themselves. Statistics will, and should, almost always be used to illustrate a relationship. It’s more important for people to remember the relationship than the number.”

Presenting dashboards by themselves can be great, but people are so overwhelmed with everything going on in their day and a myriad of data and facts being thrown around that they may gloss over your findings despite the amount of time you have put into making readable and insightful charts. You can drive home insights faster and better by tying your charts to a personal experience, a hypothetical data story, or something going on in the company. In fact, some of the best public speakers almost exclusively use stories to illustrate their points.

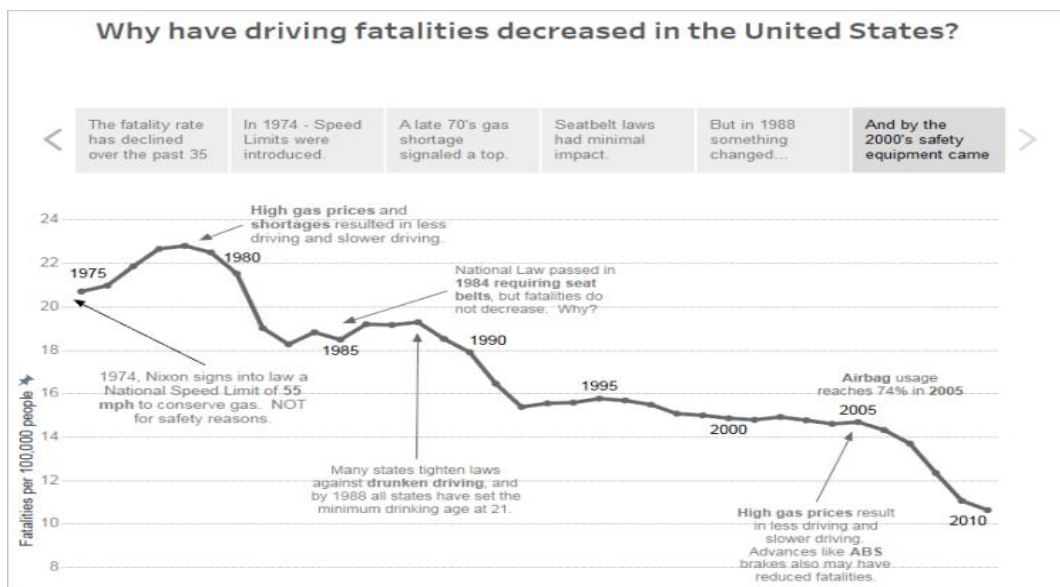
Telling a story with data and analytics can also bring your users through your thought process so that they arrive at the same conclusion that you did.

Every Story Point contains one Worksheet or Dashboard. Selections (Filtering, etc) made in the Story point are saved specifically to that story point, and do not apply to other Story Points. Since you can select one Worksheet for one Story point and another for the next, they are highly flexible.

Best Practices Tableau: https://help.tableau.com/current/pro/desktop/en-us/story_best_practices.htm



Department of Computer Science and Engineering (Data Science)



An Example Data Story (Single Story Point).

Lab Assignment:

Dataset Download:

<https://data.world/annjackson/world-indicators-tableau>

Alternate Dataset Link:

<https://drive.google.com/file/d/13-O1LoZzCSqFsHNit-u60xH1IOO8Lp9X/view?usp=sharing>

Create an Interactive Data Story with the World Indicators. The Data Story should Highlight effects of world events. World Indicators contain Information on Health (Life Expectancy, Infant Mortality, etc), Business, Development and Much More. The data pertains mainly to the 2000's.

For Example: An Analysis on African Countries affected by Wars, and how it affected the Economy, Mortality. E.G: Sierra Leone Civil War, The Congo War, Libya, South Sudan, etc. (Google can be helpful to get a list of such events!)

Or: Analyse the correlation between the ease of business and the GDP per Capita.

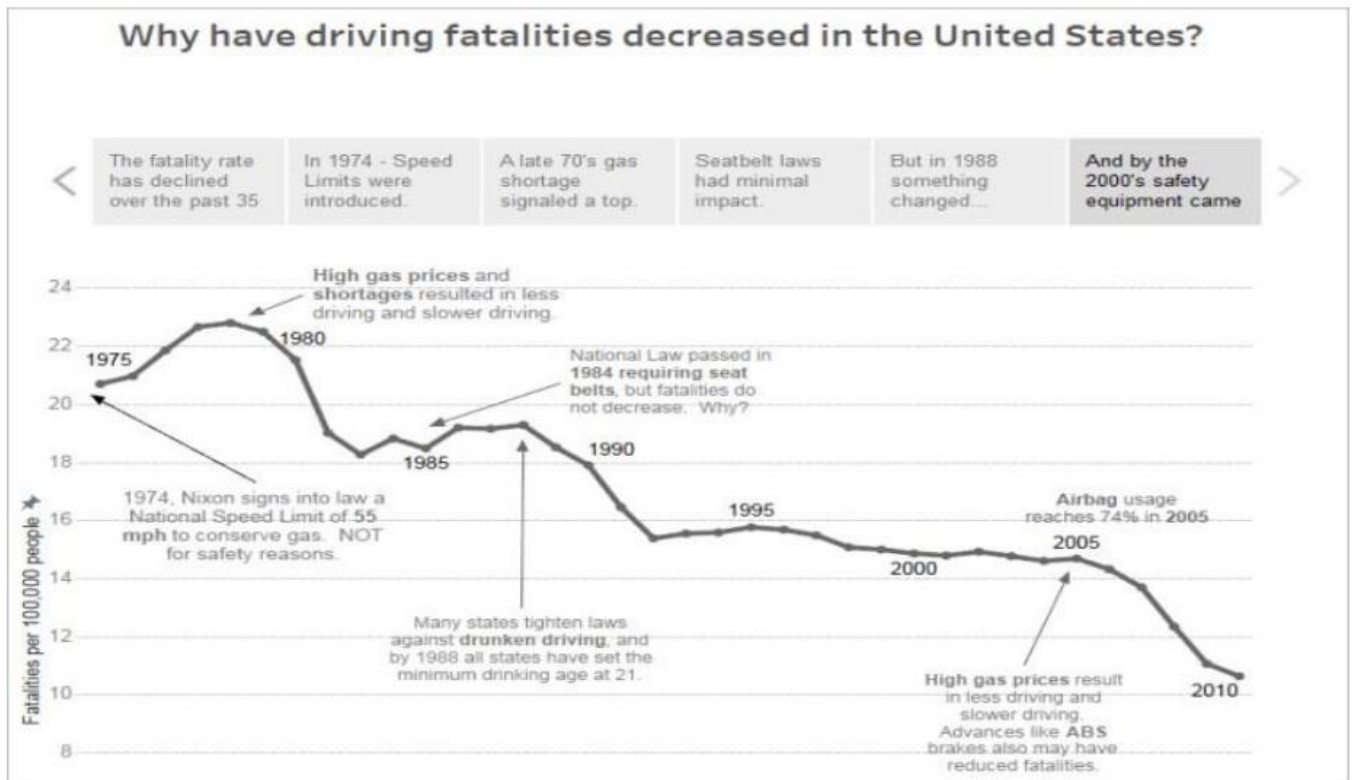
Stories will be graded based on Originality of Work, Originality of Idea, and final means of conveying your idea.

It may be worth working on this one a bit extra, as it can be a great talking point during Internship interviews, etc.

Experiment-6

Bhuvi Ghosh

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Q) Create an Interactive Data Story with the World Indicators. The Data Story should Highlight effects of world events. World Indicators contain Information on Health (Life Expectancy, Infant Mortality, etc), Business, Development and Much More. The data pertains mainly to the 2000's. Stories will be graded based on Originality of Work, Originality of Idea, and final means of conveying your idea. It may be worth working on this one a bit extra, as it can be a great talking point during Internship interviews, etc.

Storyline: Analysis of African countries in terms of health & economy by considering parameters like GDP, inbound & outbound tourism, male & female expectancy rate & infant mortality rate in a span of 12 years, interpreting the trends if any & understanding the causes behind the visualised results.

1) Analysing in terms of economy on the basis of GDP, inbound & outbound tourism & business tax rate.

Story 1

Storyline: The given storyline measures economic growth in terms of GDP, inbound & outbound tourism & business..

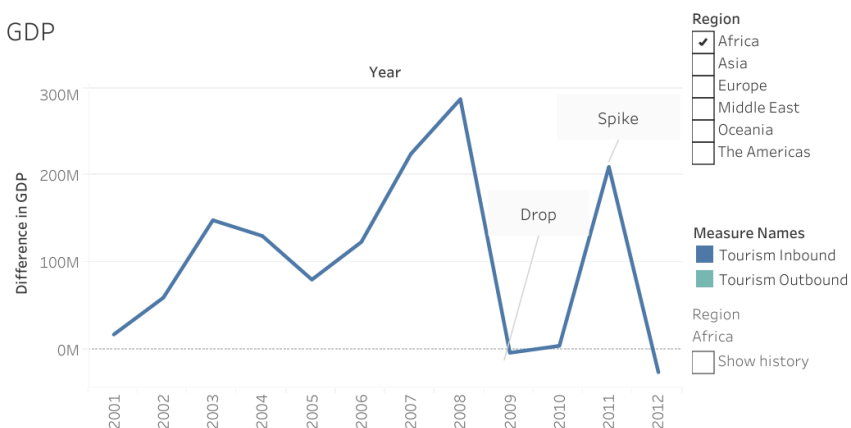
There is a drop in GDP in the year 2009 & spike in 2011.

There is a sudden drop in the inbound tourism in the year 2009 whereas outbound tourism is increasing continu..

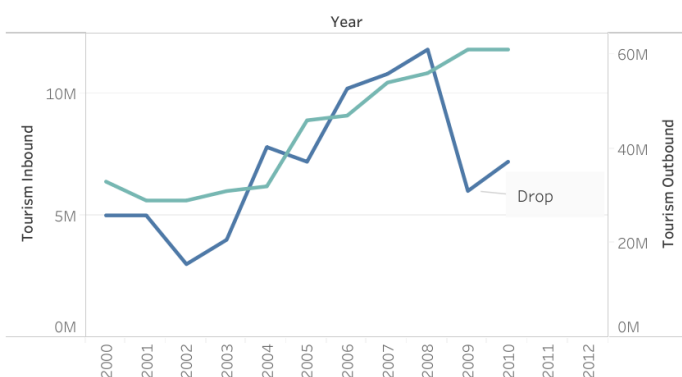
African continent



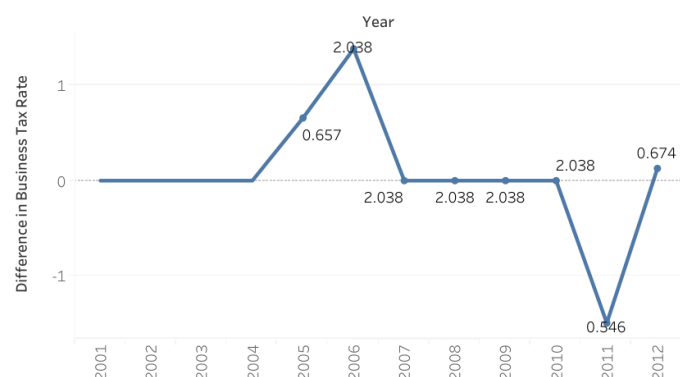
GDP



Inbound & outbound tourism



Business Tax Rate



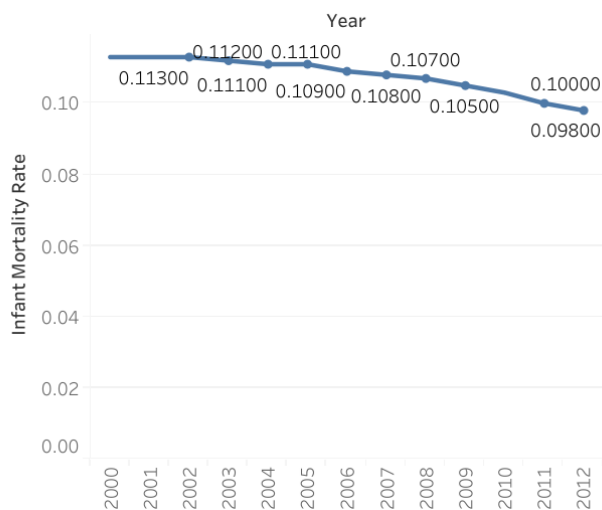
2) Analysis of health in terms of life expectancy rate of males & females & infant mortality rate.

Story 3

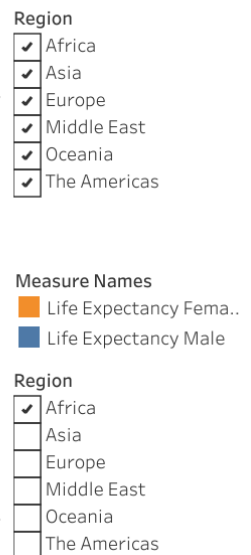
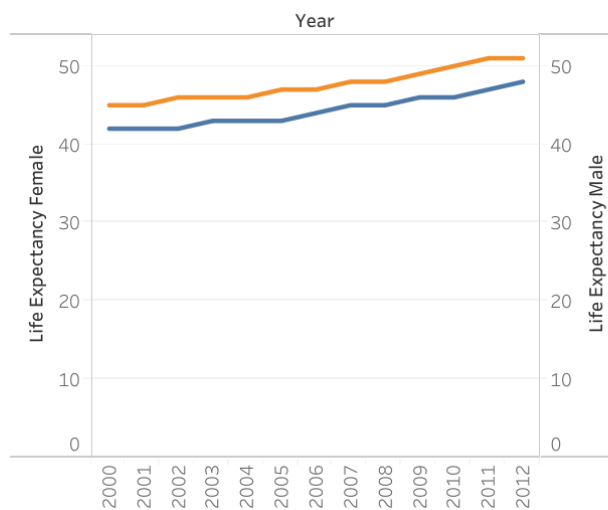
Storyline: Analysis of health in terms of life expectancy rate for males & females & infa..

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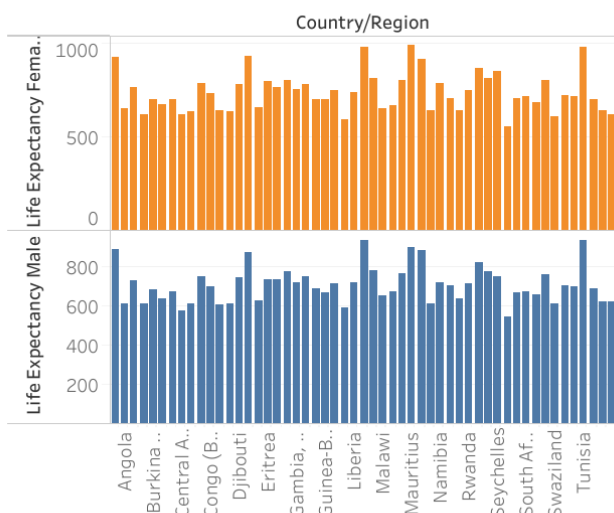
Infant Mortality Rate



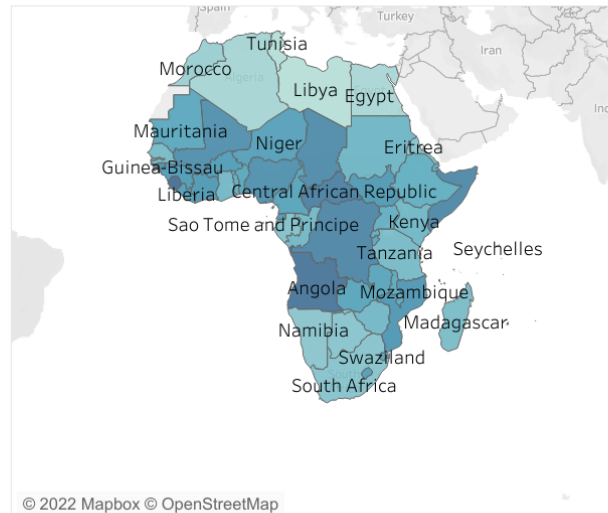
Trends in life expectancy rate



Life expectancy rate



Infant Mortality Rate



Storyline: Analysis of African countries in terms of health & economy by considering parameters like GDP, inbound & outbound tourism, male & female expectancy rate & infant mortality rate in a span of 12 years, interpreting the trends if any & understanding the causes behind the visualised results.



Storyline: Countrywise & yearwise analysis is made in Africa to analyse health & economy.

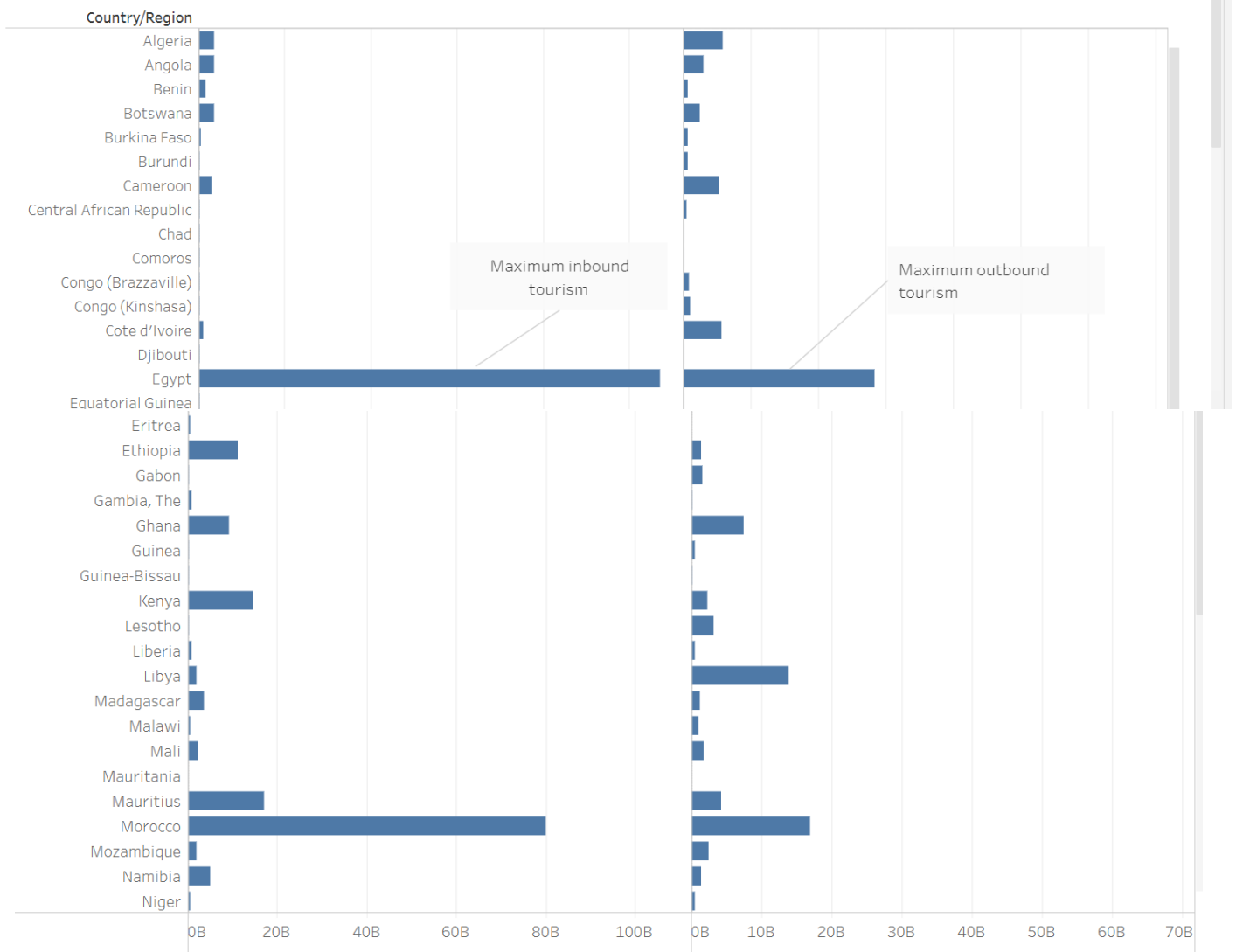
Countrywise analysis of inbound & outbound tourism shows that Egypt has the maximum revenue generation due to

South Africa followed by Nigeria & Egypt have the maximum GDP.



Countrywise analysis of inbound & outbound tourism:

< Storyline: Countrywise & yearwise analysis is made in Africa to analyse health & economy. Countrywise analysis of inbound & outbound tourism shows that Egypt has the maximum revenue generation due to South Africa followed by Nigeria & Egypt have the maximum GDP. >

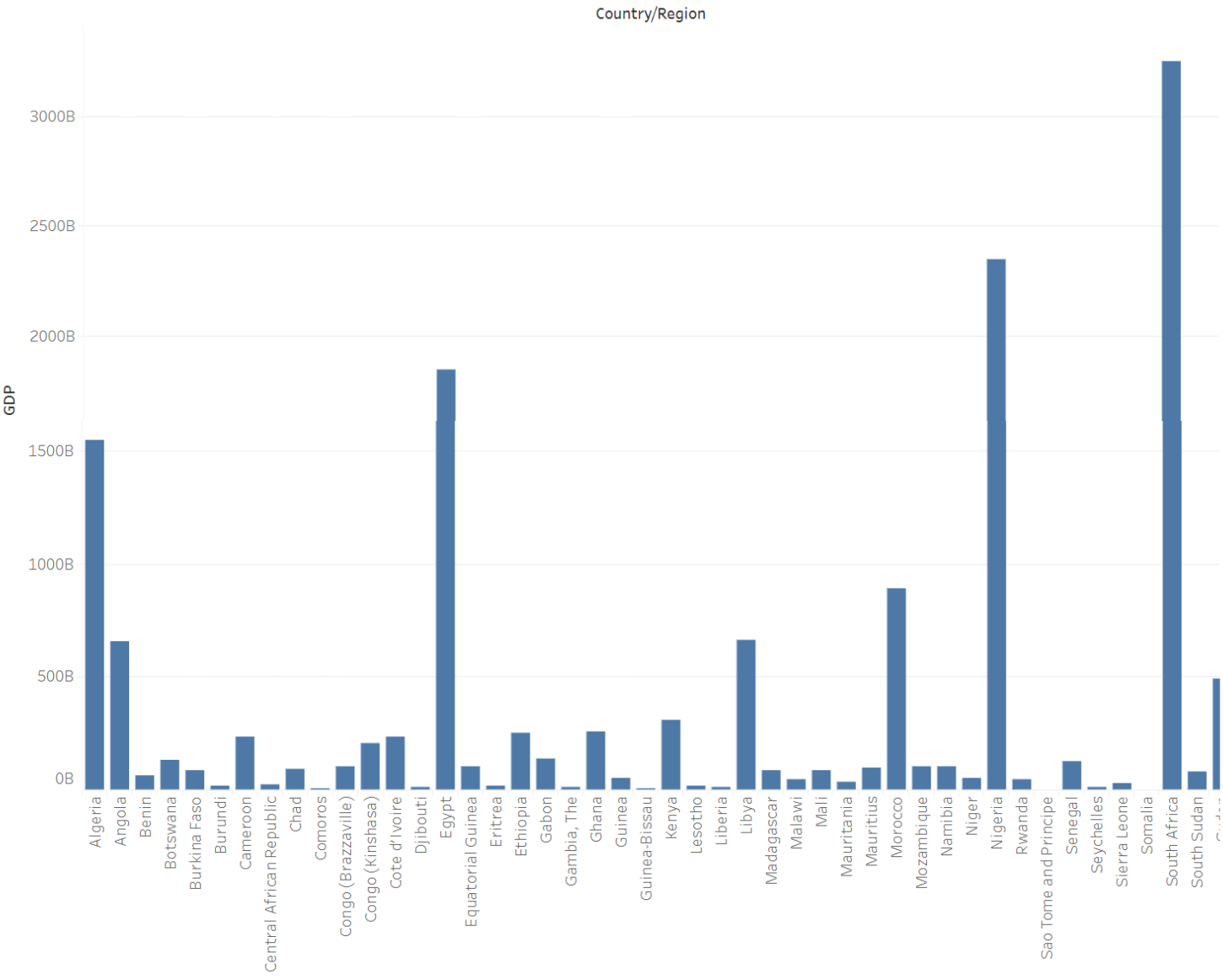


Countrywise analysis of GDP:

Storyline:Countrywise & yearwise analysis is made in Africa to analyse health & economy.

Countrywise analysis of inbound & outbound tourism shows that Egypt has the maximum revenue generation due to

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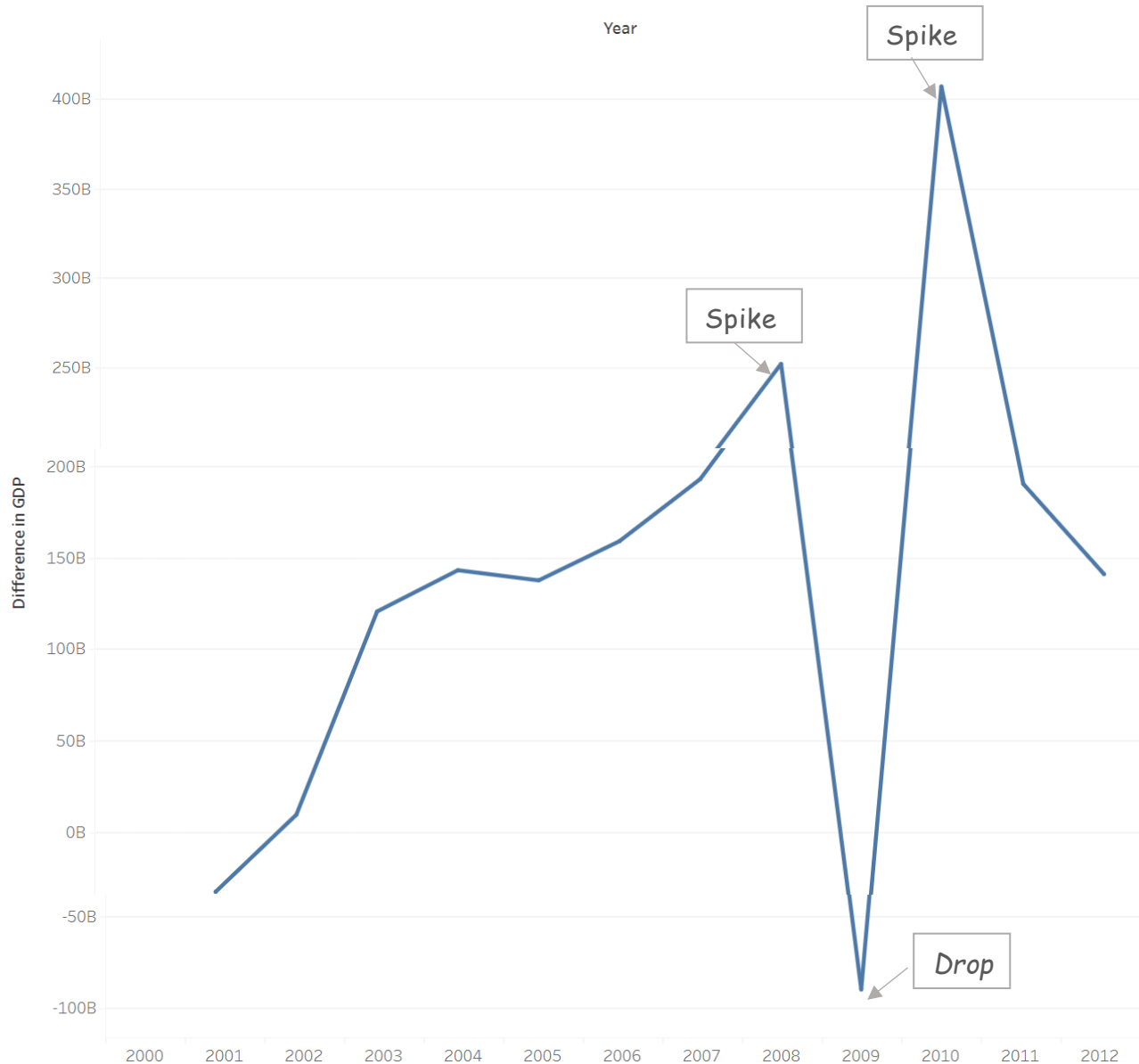


Analysis of GDP in the form of a time series data:

< Analysis of GDP shows a spike in mid 2010 & a sudden drop in 2009. The drop can be because of wars like the Syrian wars & famine like conditions.

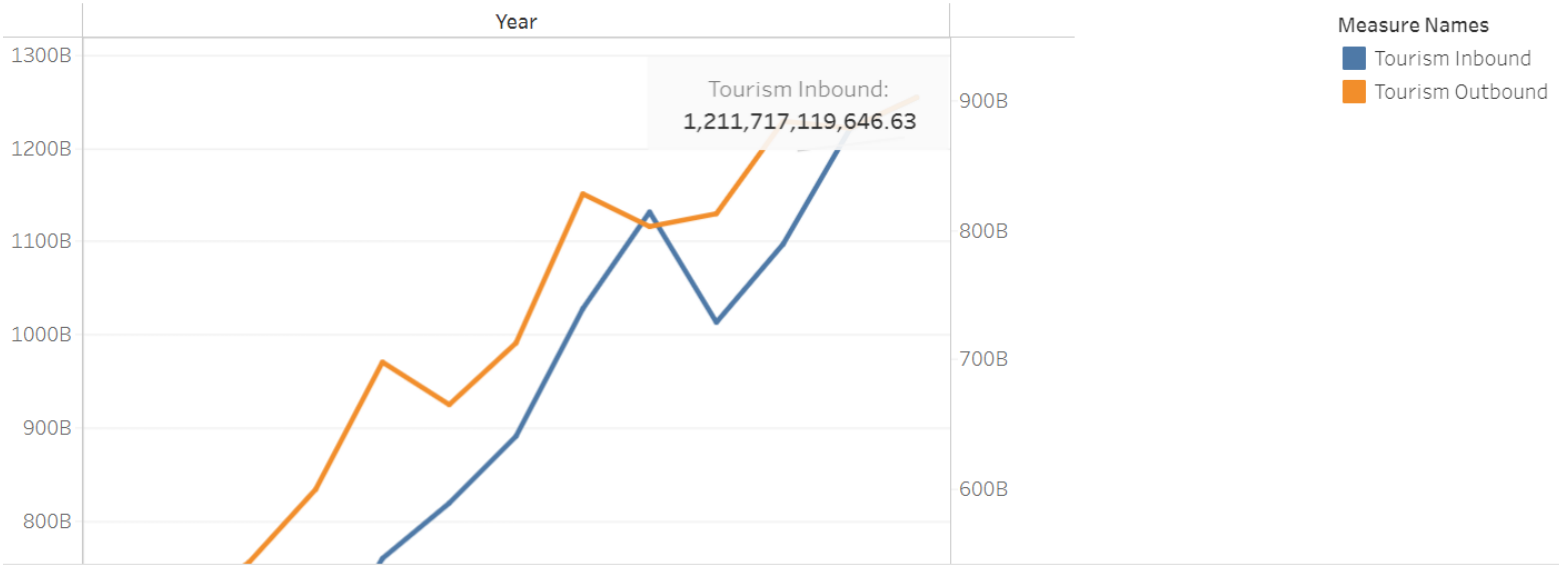
Analysis of inbound & outbound tourism shows that there is a steady increase in both inbound & outbound tourism.

Add a caption

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Analysis of trends in inbound & outbound tourism:

Analysis of GDP shows a spike in mid 2010 & a sudden drop in 2009.The drop can be beacuse of wars like the Syberian wars & famine like conditions.	Analysis of inbound & outbound tourism shows that there is a steady increase in both inbound & outbound tourism.	Add a caption
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Change in the male & female expectancy rate over the span of 12 years:



mid 2010 & a sudden drop
of wars like the Syberian

Analysis of inbound & outbound tourism shows that there is
a steady increase in both inbound & outbound tourism.

There is a spike in life expectancy of both males & females
in the year 2000 which can be attributed to improvement
in healthcare facilities.

Add a

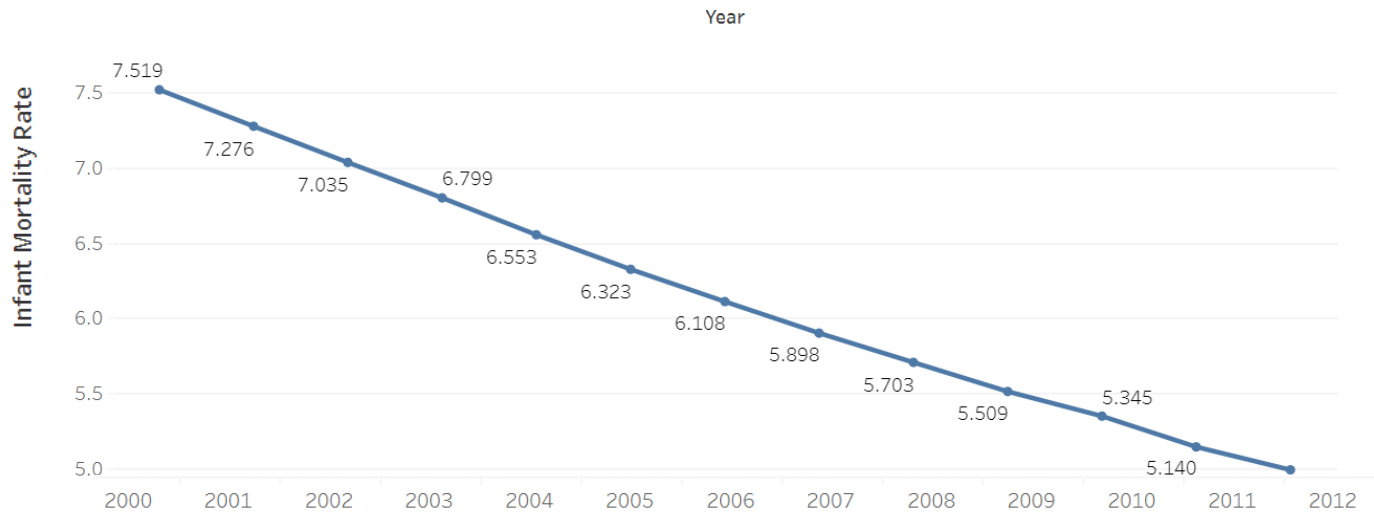


Infant mortality rate:

Decreasing infant mortality rate is observed in Africa which can be attributed to improvement in healthcare facilities.

Decreasing infant mortality rate is observed in Africa which can be attributed to improvement in healthcare facilities.

Add a caption



Countrywise analysis of infant mortality rate:

